Heitaria Swiss AG
ASSET MANAGEMENT

TECHNICAL REPORT · PHASE 0

SALAR DE UYUNI — NOR LÍPEZ, POTOSÍ

Hydrological and Hydrogeological Assessment

Phase 0 — desk study, prior to drilling

PROJECT	Lithium carbonate, 80,000 t/yr plus a 10,000 t/yr pilot plant, from brine of the lithium-rich zone of Nor Lípez. Route C flowsheet, without evaporation ponds.
SUBJECT	Brine extraction of 12.23 Mm³/yr from YLB's existing wellfield, and its consequences for the brine table, the brine/freshwater interface, the project water balance and the disposal of two reject streams.
TECHNICAL BASIS	Techno-economic model (current sections), site master plan and published literature on the Salar de Uyuni. No field data have been obtained for this project.
RECIPIENT	CUPCONL — Central Única Provincial de Comunidades Originarias de Nor Lípez (the provincial union of indigenous communities of Nor Lípez), and the technical advisers it designates.

Version 1.7 — 12 November 2026. Updates the commitments on the page "The project"; no change to the hydrogeological figures, methods or conclusions. Version 1.6 — 7 October 2026. Adds Annex A, on what European battery buyers will ask about water under the EU Batteries Regulation; no change to the figures, the methods or the conclusions. Version 1.5 — 26 August 2026. Copy prepared for CUPCONL. Supersedes version 1.4 with a single addition and no change whatsoever to the figures, the methods or the hydrogeological conclusions: it adds the page "The project", which describes what project this is — production, process route, employment, commitments and fiscal position — so that the document does not assume the reader arrives with that context. Version 1.4 had redrawn Figure 2, whose circles were being traced over ground where no brine aquifer exists (Section 4.3); restated make-up water as a structural condition (Sections 5.1 and 5.2); corrected the effect of density and viscosity on transmissivity (Section 4.4); and corrected the statement of assumptions for the Thiem solution (Section 4.1).

This is a desk study. It is not an authorized hydrogeological study and does not carry the signature of a professional registered with RENCA. No field data have been obtained for this project, so its quantitative results are order-of-magnitude and are flagged as such throughout the document. Section 8 sets out the measurement program that will replace them.

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RESPONSES AND ENQUIRIES

Comments from the advisers designated by CUPCONL may be addressed in writing to this address and will be answered in writing.

EXECUTIVE SUMMARY

Status of this assessment

Every figure in this document comes directly from the project's own techno-economic model, is derived arithmetically from it, or comes from published literature on the Salar de Uyuni. Nothing has been measured in the field, because no field program has yet been carried out.

That limits what can be concluded, but less than might be expected. The extraction volume is fully determined by production, grade and recovery, and all three are on paper. What that volume does to the brine table depends on aquifer parameters that have not been measured — but the arithmetic can be run across the entire plausible range of parameters, and wherever the answer is insensitive to that range it constitutes a result, not an assumption.

Purpose of this copy. This document is a technical assessment, delivered to CUPCONL so that the communities of Nor Lípez and their own advisers can examine the project's water figures directly, and not through a summary. It is not a consultation instrument and does not substitute for free, prior and informed consultation. It carries no request for a decision.

It is delivered in the same form in which it is used internally. Every method, every assumption and every unresolved gap is stated, including those that count against the project. Sections 7 and 8 list what is not yet known and what will be measured to close it. Where CUPCONL's advisers disagree with a parameter, the arithmetic is laid out in detail so that another value can be substituted and the consequence followed.

The project undertakes to share with CUPCONL the results of the Section 8 field program as they are produced, and not only in the eventual EEIA (environmental impact study).

12.23 Mm³/yr	0 m³/yr	27 to 42 km	80,000 t/yr
BASE-CASE EXTRACTION	FRESHWATER CONSUMED IN OPERATION	RADIUS OF INFLUENCE CENTRAL BAND	Li_2CO_3 PRODUCTION

Full parameter range: 13 to 94 km. To the south the salar margin truncates the cone at about 13 km from the wellfield. Section 4.3.

CRITICAL SYNTHESIS

The four most important results

The cone of depression most plausibly reaches 13 to 94 km — an analytical range. It is a cone in brine, so what it defines is not whose water is affected, but three other things: that the project's extraction and YLB's share a single cone, because they are the same wells; whether it reaches the dilution zone of the delta and degrades the produced grade; and whether lowering a brine table that today sits at the surface alters the crust and the seasonal flooding that everything built on it must withstand.

The project does not consume water — it produces it. Condensate exceeds process demand by about 8,000 m³ per day, and by about 3,800 m³ per day even if process demand turns out to be the pessimistic figure. After the camp, the community commitment and the cleaning of the solar panels, thousands of cubic meters per day remain uncommitted. This is a water resource, not a water impact.

The solids are placed on the surface of the salar, in YLB's designated disposal area, 30 km away. At a thickness the crust can reasonably support, 43 years of production require approximately 30 to 100 km². Neither the extent of that area nor the bearing capacity of the crust at that location appears in any source. Section 6.1.

The magnesium reject brine — 1.467 Mm³/yr, 63 Mm³ over the life of the project — returns underground through some five to twelve injection wells. The open questions are where to locate them, so that the returned magnesium cannot reach the production zone, and under what instrument it is licensed: DS 4180 provides for reinjection into the salar in its Article 15, but conditions it on a quality equal to or better than that of the well brine, which this liquor does not meet on a literal reading. It should be raised with the MMAyA from now.

DESCRIPTION

The project

This assessment deals with one component of a larger project, and a reader who encounters 12.23 Mm³/yr of extraction before encountering the project lacks the context to judge it. This page provides that context. It is description, not assessment: its figures come from parts of the model other than those that support the rest of the document, they are identified in Section 9.1, and where they are uncertain this is said.

ITEM	VALUE
Product	Lithium carbonate (Li_2CO_3).
Steady-state production	80,000 t/yr, plus a 10,000 t/yr pilot plant that precedes it.
Resource	Brine from the lithium-rich zone of Nor Lípez: 12.23 Mm³/yr extracted from YLB's existing wellfield, at a design grade of 1.5 g/L lithium and 82% recovery. The project opens no new wellfield.
Process route	Route C, **without evaporation ponds**: concentration is by mechanical evaporation and controlled crystallization, in a US$278.4 million system that replaces the ponds. It is this decision that makes the project a net producer of water and, at the same time, energy-intensive.
Energy	Self-sufficient supply, with no grid connection: photovoltaic field with batteries and a solar thermal field. It is assessed in a separate document, and is the reason why panel cleaning appears in the water balance of Section 5.
Phases	Pilot plant built in year 1, tested in year 2 and commercial from year 3. Main plant built between years 3 and 5, commercial in year 6.
Modeled life	45 years, of which 43 in production.
Total cost	US$3,465.4 million.
Structure	Bolivia 51% and sponsor 49%. The sponsor contributes 100% of the plant's equity; Bolivia contributes no cash.
What it takes from YLB	A single element: the use of the salt disposal cells, at no cost. Brine, water and energy are self-supplied.

Employment and presence in the territory

PERMANENT POSITIONS IN OPERATION	PEOPLE	CONTENT
Operations	390	Continuous 80,000 t/yr plant, four-shift roster.
Maintenance	187	Mechanical, electrical and instrumentation.
Technical, laboratory and administration	125	Metallurgy, quality control, health and safety, finance.
Total permanent in operation	702	Rotation regime, with about 12 rotations per year and transport to Uyuni and Oruro.

PERMANENT POSITIONS IN OPERATION	PEOPLE	CONTENT
Construction peak	**≈1,400**	**Estimated workforce at the peak of construction, between years 1 and 5. It is the basis on which the residential camp and its water plant were sized — see Section 5.**

The two figures describe different phases and must not be added together. The construction peak is transitory and larger; the 702 positions are permanent and are the ones that remain over the 43 years of production.

> **The gap worth stating.** The 1,400 people are a peak value and not a curve: neither the pace of hiring over the course of construction nor the proportion of local hiring appears anywhere in the model. For the communities of Nor Lípez, that proportion is probably the most significant figure in the project, and it is a legitimate question that this document cannot answer.

Commitments and fiscal position

ITEM	VALUE	NOTE
Río Grande package	US$2.95m/yr	Impact and land use. Recurring operating cost, CPI-indexed.
Community program	US$7.8m/yr	Host territory and municipalities. Recurring operating cost.
15% program	~US$42.6m/yr	15% of the profits of the private 49% share, for Potosí. Average year at US$15,000/t; zero in a loss year. Additional to the items above. With training and workforce transport: about US$53.9m/yr in total.
Water supply to communities	To be decided	The current contractual commitment is the 200 m³/day to Río Grande (Section 5.3). A broader distribution program was costed and is currently switched off in the model; the surplus that would make it possible exists and is quantified in Section 5.4.
Current royalty regime	Exempt	Industrialized lithium carbonate, Article 224 of Law 535.
Draft lithium law	**3 to 10%**	**The technical working groups of July 2026 would remove lithium from Law 535 and create a progressive royalty on the international price, in bands from US$9,000 to US$35,000/t. At the base price of US$15,000/t it is equivalent to 4.6%.**
Profits tax	32.5%	IUE of 25% plus 7.5% on value added.

The model runs with the draft law's royalty applied, and not with the current exemption. It is the conservative reading. A legal opinion on transition and acquired rights remains pending, which the model flags as necessary and which is not part of this assessment.

> Nothing on this page forms part of the hydrogeological assessment. It is included because a document delivered to the communities of Nor Lípez cannot assume that the reader arrives with the project context, and because several of the figures above — the proportion of local hiring, the scope of the water supply to communities, the transition of the royalty regime — are open, and it is preferable that they be read as open rather than as omitted.

TECHNICAL MATRIX

Key questions and preliminary answers

QUESTION	ANSWER AT THIS STAGE	BASIS
How much brine must be extracted?	12.23 Mm³/yr in the base case. Working range of 8 to 27 Mm³/yr.	Derived from production, grade and recovery. Section 2.
Does the project extract freshwater?	No. None, in any operating case. It extracts brine only.	The water balance closes on condensate alone, and with a surplus. Section 5.
Does it take water that Río Grande uses?	No. The project extracts brine only and takes no freshwater in any operating case. Río Grande is supplied from the San Gerónimo aquifer through wells and springs; the relationship with brine drawdown is set out in Section 3.3 and its magnitude remains to be measured.	Section 3.3.
Can the community commitment of 200 m³/day be met?	Yes. It amounts to 2.4% of the surplus. The exposure is in continuity, not in quantity.	Section 5.3.
Could the project supply more than the committed 200 m³/day?	Physically yes. Of the order of 3,150 m³/day would remain available even with a pessimistic process demand. The project will also establish a collection point at the plant, open to all the communities of Nor Lípez; the contractual volume commitment remains the 200 m³/day to Río Grande. Section 5.4.1.	Section 5.4.
Does the project open a new wellfield?	No. It extracts from YLB's existing wells, within an already licensed volume. No new point of extraction is created on the salar.	YLB's own environmental filing. Section 2.4.
How far will the BRINE table fall?	Of the order of 4 to 14 m near the wellfield and 1 to 5 m at 10–20 km with the project's own extraction; 2.40 times those figures at the upper bound of 29.35 Mm³/yr.	Thiem and volumetric analysis across the full parameter range. Section 4.4.
Is there freshwater beneath the salar?	No. The pore fluid is brine throughout the profile — from the crust to depth, across the whole interior. Nowhere in this assessment is a freshwater level predicted.	Section 3.3.
Is there any link between extraction and the river?	One, and it operates in the reverse direction: a lower brine table causes the river to lose marginally more to infiltration in its delta. It is second-order and is measured against the river's flow.	Section 4.6.
How far does the cone of depression reach?	It is a cone in brine: what falls is the brine table beneath the salar, not any freshwater. Depending on the aquifer parameters, some lowering of that brine table could be measurable 13 to 94 km from the wellfield after 43 years. But the brine aquifer ends at the edge of the salar — about 13 km to the south — so in most cases the cone simply reaches the margin, and the 94 km upper end is a mathematical limit that would extend far beyond the salar and cannot physically occur. The useful measure is therefore not the radius but how much the brine table falls at given places: about 4 to 14 m near the wellfield and 1.6 to 3.8 m at the southern margin, at the project's own extraction rate.	Radius-of-influence analysis, truncated at the salar margin. Sections 4.3 and 4.4.
Can extraction dilute the brine the project produces?	Possibly, over decades, if the cone extends toward the dilution zone of the delta. It is a grade risk for the project, not a risk to anyone's water.	Section 4.5.
How much area does solids disposal require?	88 to 113 Mm³ of volume; the area depends on placement thickness. The capacity of the designated area is not documented.	Section 6.1.

QUESTION	ANSWER AT THIS STAGE	BASIS
Where does the magnesium reject brine go?	Back underground, through some five to twelve injection wells. Their location must keep the returned magnesium away from the production zone. The obstacle is regulatory: DS 4180 provides for reinjection into the salar in its Article 15, but under a quality condition that this liquor does not meet.	Section 6.2.

STRUCTURE

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03	**Preliminary conceptual model**	Hydrostratigraphy, parameters and the San Gerónimo aquifer
04	**Anticipated effects of extraction**	Cone of depression, analytical drawdown and dilution
05	**Project water balance**	Water self-sufficiency and surplus for communities
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A	**European battery buyers**	What their water due diligence will ask, and where this assessment answers

SECTION 01

1. Physical setting

1.1 The salar

The Salar de Uyuni occupies some 10,000 km² of the southern Altiplano, at about 3,650 m. It is a closed basin: water leaves by evaporation, not by runoff. Its surface is a halite crust of the order of a few meters to about ten meters thick, underlain by alternating lacustrine sediments and other saline units, with brine occupying the pore space throughout the profile.

The thickness of the crust matters to this project for two distinct reasons. It is the founding medium for the conveyor piles, which the model specifies at 6 to 12 m — that is, piles that pass through the crust and penetrate into whatever lies beneath. And it is the uppermost unit of the aquifer from which the brine will be extracted.



Figure 1 — Project location. The plant site on the southern margin of the salar, the brine pipeline and the wellfield to the north, and the solids conveyor corridor toward YLB's disposal area. Satellite base: Esri / Maxar.

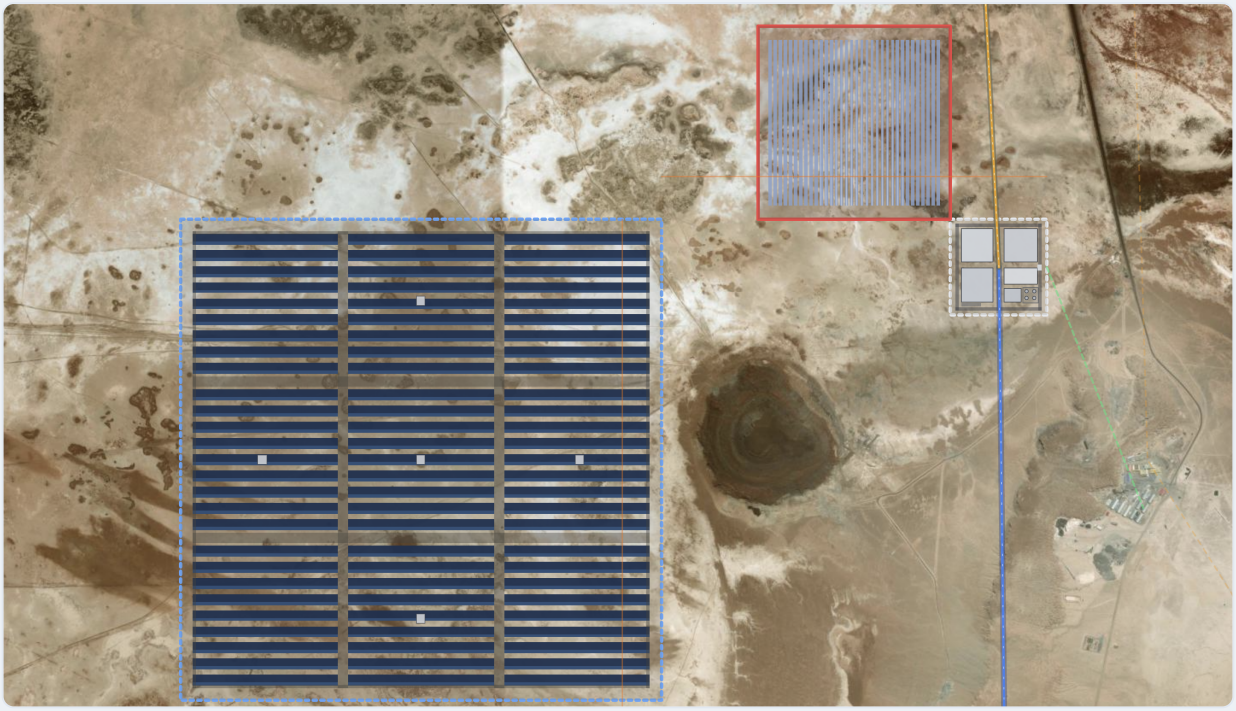


Figure 1b — Detail of the plant site: photovoltaic field (2.5 × 2.5 km), solar thermal field (1 × 1 km) and process plant, with Llipi Camp to the east.

1.2 Climate and natural water balance

PARAMETER	VALUE	SOURCE AND RELEVANCE
Gross evaporation	1,500 mm/yr	Model. Approximately half the Atacama rate — the reason why evaporation-based designs developed for Chile fail here.
Precipitation	350 mm/yr	Model. More than twenty times the Atacama figure, concentrated between December and March.
Brine evaporation factor	0.65	Model. Brine evaporates more slowly than freshwater.
Net evaporation over brine	0.7475 m/yr	Derived in the model. The natural rate at which the salar loses brine to the atmosphere, and the reason why a closed basin at this altitude retains salt rather than water.
Rainy season	December to March	Four months in which evaporation is practically nil and the salar margin floods.
Winter air temperature	−10 to +10 °C	Published literature on the salar. The daily range, not a sustained value.
Summer air temperature	+10 to +25 °C	Published literature on the salar. Evaporation at Uyuni is approximately half the Atacama rate, largely because of these lower temperatures — which is why designs developed for Chile are not transferable.

One consequence is worth pointing out from the outset: the rainy season, December to March, is the constraint that governs almost every part of this project — and all of those demands fall in the same window, not one at a time. The salar margin floods, which is why the solids conveyor is elevated on piles above the flood level and why truck haulage was ruled out from the start. Some 350 mm of rain fall directly on the disposal area, dissolving the soluble fraction of the solids at precisely the one time of year when the surface

is covered by standing water. Solar generation is at its worst: every multi-day cloud and snow event falls within this window, which is precisely the reason for the emergency generation — some 145 operating hours per year, supplied in advance and not delivered over the course of the season. The rainy season is also the warm season, so the winter conditions on which the process depends for part of its chemistry are not available throughout that period.

One coincidence operates in the reverse direction, and it is the one that matters most to the communities. The plant produces water by evaporating brine with solar energy, so condensate production peaks in the dry season — which is exactly when the Río Grande de Lípez is at its minimum. The supply the project provides is greatest when the river the community depends on is weakest. In terms of water, the two seasons complement rather than compound each other, and that is worth saying explicitly rather than leaving it to be noticed.

The practical consequence throughout this assessment is that the hydrological year, and not the calendar year, is the correct frame. Annual averages hide the problem: a figure that works over twelve months can fail in the four that matter.

1.3 The Río Grande de Lípez

The Río Grande de Lípez enters the salar from the south and is the main surface inflow to this part of the basin. Its role in this project is limited and specific, and it is worth being precise because it is easily overstated.

- It is not a water source. The project extracts neither surface water nor fresh groundwater.
- Its delta is a freshwater inflow to the southern margin of the salar and therefore constitutes a boundary condition for any groundwater model — it is one of the controls on where the brine/freshwater interface sits.
- Its wetland is a physical obstacle. The solids conveyor is routed over 38 km rather than the 30 km straight-line distance precisely to go around it.

SECTION 02

2. Extraction demand

2.1 Derivation

Extraction is not a design decision. It is fixed by how much lithium the plant must sell, how much lithium each cubic meter of brine carries, and how much of that lithium survives the process.

> The 10,000 t/yr pilot plant is not included in this derivation: it operates as a precursor and ceases when the main plant comes on line (if it operated in parallel with the main plant, extraction would rise to 13.76 Mm³/yr and all the figures in Sections 4 and 6 would scale by 1.125).

TERM	VALUE	NOTE
Production	80,000 t/yr Li_2CO_3	Project target. Fixed.
Lithium content of Li_2CO_3	18.788%	Stoichiometric: 2 × 6.941 / 73.891.
Lithium in product	15,030 t/yr	Production × content.
Lithium per tonne of product	188 kg	The model's working figure.
Brine grade	1.5 g/L	Adjustable variable, provisional value. Nor Lípez rich zone documented between 1.5 and more than 2.0 g/L. Whole-salar average 0.42 g/L.
Overall recovery	82%	Controlled crystallization, Route C. Parametric — no vendor quotation.
Brine per tonne of product	152.8 m³/t	188 / (grade × recovery).
EXTRACTION	**12.23 Mm³/yr**	≈ 33,500 m³/day, ≈ 1,396 m³/h continuous.
Over 43 years of production	526 Mm³	The figure that governs Section 4.

2.2 Sensitivity

Grade and recovery are both provisional values. Neither has been measured on this project's brine. Their product sits in the denominator, so the effect on extraction is more than linear in either one taken separately.

CASE	GRADE	RECOVERY	BRINE PER TONNE	EXTRACTION
Highest documented grade	2.3 g/L	82%	99.7 m³/t	7.97 Mm³/yr
Upper bound of the rich zone	2.0 g/L	82%	114.6 m³/t	9.17 Mm³/yr
MODEL BASE	**1.5 g/L**	**82%**	**152.8 m³/t**	**12.23 Mm³/yr**
Base grade, reduced recovery	1.5 g/L	55%	227.9 m³/t	18.23 Mm³/yr
Low grade and low recovery	1.0 g/L	55%	341.8 m³/t	27.35 Mm³/yr
Whole-salar average grade	0.42 g/L	55%	813.9 m³/t	65.11 Mm³/yr

The working range is 8 to 27 Mm³/yr. The last row uses the whole-salar volumetric average grade of 0.42 g/L, derived from Risacher and Rettig (349 mg/kg; 424 mg/L at density 1.21). CEDIB's own figure (250 ppm, ≈0.30 g/L) is lower still, so the lower bound assessed here is conservative against that source as well. It is included because it shows what is at stake in the grade question: if the wellfield turned out to be at the salar average and not at the rich-zone grade, extraction would be five times the base case and this would be a different project. It is a bound of the sensitivity analysis, not a description of this wellfield — Section 2.4.1 explains why.

TECHNICAL ANALYSIS

What drilling actually resolves

The model is explicit that grade is not a bankability lever: the DSCR moves only from 1.12 to 1.18 across the entire documented grade range, and the covenant is met by price and capital structure, not by grade.

But grade determines brine volume, and brine volume is the entire subject of this assessment. A result of 1.5 g/L and one of 2.0 g/L differ by 3 Mm³ per year — roughly a quarter of the base extraction — and everything in Sections 4 and 6 scales with it.

Drilling answers a hydrogeological question more than a financial one. That is a reason to run the confirmation program and the hydraulic program as a single campaign, and not as two.

A cross-check against the source of the flowsheet — and what it does and does not say. The published basis for this flowsheet works a southern Uyuni brine of about 2.7 g/L and derives 68 m³ of brine per tonne of Li_2CO_3, explicitly without process losses. The model derives 152.8 m³/t. The difference is 2.25 times against the published 68 m³/t (or 2.19 times against the 69.6 m³/t of the formula itself), and it decomposes cleanly: 1.80 for using a lower grade, 1.22 for applying a real recovery instead of none.

What this means is that the assessment is built on a more demanding case than the published chemistry. It does NOT mean that the plant extracts twice the brine it needs. Every impact in this document scales with extraction, so if the wellfield delivers something closer to the published grade, the actual figures will fall below those assessed.

The distinction is stated explicitly because the phrase "twice the published basis" can be read in two ways: the larger figure comes from assuming a leaner brine and applying real process losses, not from a plant that uses twice the brine it needs.

What the assessed figure would become if drilling delivers a grade within the documented Nor Lípez range:

CASE	BRINE PER TONNE	EXTRACTION	AGAINST THE 12.23 MM³/YR ASSESSED
Assessed base — 1.5 g/L, 82%	152.8 m³/t	12.23 Mm³/yr	100% — every figure in this document
Drilling delivers 2.0 g/L	114.6 m³/t	9.17 Mm³/yr	75%
Drilling delivers 2.3 g/L (highest documented grade in Nor Lípez)	99.7 m³/t	7.97 Mm³/yr	65%
Published southern grade, real recovery	85.0 m³/t	6.80 Mm³/yr	56%

The practical consequence: the drawdown, the cone radius, the solids tonnage and the reject brine volume in this assessment are all upper estimates as far as the grade question is concerned. A better drilling result creates no new problem to assess — it reduces every figure already assessed. The reverse case, a wellfield below 1.5 g/L, is the one that would require this work to be redone, and is the reason why confirmation drilling matters more to the hydrogeology than to the financial model.

2.3 What the extracted brine carries

Every cubic meter of brine brings dissolved salts that must leave the site as solids or as liquor. At base-case extraction, the published salt yields give the following annual tonnages, all of which scale directly with extraction.

CONSTITUENT	YIELD	AT 12.23 MM³/YR	DESTINATION
Halite (NaCl)	212.5 kg/m³	2,598,374 t/yr	Disposal area, via the 38 km conveyor.
Carnallite ($KMgCl_3 \cdot 6H_2O$)	62.5 kg/m³	764,228 t/yr	Disposal area. Deliquescent.
Magnesium sulfate	25 kg/m³	305,691 t/yr	Disposal area.
Boric acid (H_3BO_3)	2.5 kg/m³	30,569 t/yr	Marketable product, currently booked with zero credit and therefore treated as waste.
Total crystallized	**302.5 kg/m³**	**3,698,862 t/yr**	—
Solids conveyed	—	3,668,293 t/yr	Excludes boric acid.
Magnesium reject brine	12% of brine	1,467,317 m³/yr	Conveyed by a separate pipeline. No solid bischofite forms.

Approximately 302 kg of salt crystallize from every cubic meter of brine extracted. Over 43 years that represents of the order of 159 million tonnes crystallized, of which 158 million are conveyed to the disposal area once boric acid is deducted, and 63 million cubic meters of reject brine. Sections 6.1 and 6.2 address where they go.

2.4 Extraction from an existing wellfield

The project does not open a wellfield. The brine is extracted from YLB's existing wells, which have been in operation for years and sit within an extraction volume that is already licensed and already subject to an environmental filing. That single fact governs most of what follows, and it is the most important difference between this project and a new brine operation.

	YLB, AS LICENSED	THIS PROJECT
Brine extracted	17.12 Mm³/yr — declared in its own environmental filing	12.23 Mm³/yr assessed, from the same wells — 29% of the licensed volume unused in the base case
Lithium carbonate produced	Nominal capacity 15,000 t/yr. The published review of the salar recorded no Bolivian lithium production up to 2017, and the project's own notes point out that YLB has never reached its nominal capacity with this brine	80,000 t/yr design
Potassium chloride produced	Nominal capacity 350,000 t/yr, likewise not reached	None
Pond infrastructure	160 ponds, 22,000 ha built; 48 usable — 30%. Seventy were never lined	None
Documented lithium recovery	Below 55% (CEDIB)	82% as design basis, unvalidated

Put plainly: no new wellfield is opened, no new point of extraction is created on this salar, and no area of influence is established that does not already exist. What changes is how much lithium is produced from brine whose extraction is already licensed. Whether total extraction rises, falls or stays the same depends on how much YLB continues to extract on its own account, which appears in no available source — which is why Section 4 is worked across the full range.

TECHNICAL ANALYSIS

Why the total extraction figure is presented as a range

YLB's 17.12 Mm³/yr is a licensed and declared figure, not a measured one. Its production never reached nominal capacity, so its actual extraction is unknown and is certainly lower. The project's 12.23 Mm³/yr may replace part of that extraction, or add to it.

Three cases bound the answer. If the project's extraction replaces YLB's, total extraction is 12.23 Mm³/yr — 29% below the licensed volume. If it adds to it in full, total extraction is 29.35 Mm³/yr. The real case lies between the two, and cannot be pinned down without YLB's pumping records.

Section 4 gives the drawdown at both ends of that range. The upper end is not a prediction; it is the bound that governs if YLB resumes full licensed extraction alongside this project.

2.4.1 The advantage is in the flowsheet, not in the grade

Both operations sit in the same southern rich zone — YLB's plant is at Llipi, in Nor Lípez, and the published basis for this flowsheet works a southern Uyuni brine of about 2.7 g/L. This assessment assumes 1.5 g/L, roughly 1.8 times leaner than that published figure, and still arrives at 152.8 m³/t.

So the difference does not come from better brine. It comes from a recovery of 82% against a documented figure below 55%, from not having a multi-year passage through ponds with their losses to evaporation and seepage, and from not concentrating to the point of losing lithium into carnallite. That statement does not depend on the drilling result, which makes it the most durable one.

A clarification on the whole-salar average. The figure of 0.42 g/L is not a downside case for these wells. It averages the lean northern zones of a salar 100 km across, whereas YLB's wellfield sits in the southern rich zone. It is used in Section 2.2 as the lower bound of the sensitivity range, which is where it belongs, and it is not a description of the brine these wells produce.

The cumulative case stands equally: if both operations run, the combined extraction from this salar is of the order of 29.3 Mm³/yr. No operator's assessment can treat its wellfield as isolated.

SECTION 03

3. Preliminary conceptual model

What follows is a working conceptualization assembled from published descriptions of the salar. It is not calibrated and no parameter in it has been measured on this project's ground. Its purpose is to define the parameter ranges over which Section 4 is worked.

3.1 Hydrostratigraphy

UNIT	CHARACTER	RELEVANCE
Halite crust	From a few meters to about ten meters of massive to porous halite	Founding medium for the conveyor piles and the pipeline trench. Its porosity is largely intercrystalline and its drainable fraction is the most uncertain parameter in the system.
Lacustrine sediments	Interbedded muds, silts and evaporites	Probable low-permeability layers that may separate brine horizons and sustain vertical differences in hydraulic head.
Deep halite and sediment sequence	Alternating, to considerable depth	Where the producing horizon is most likely located. Geometry unknown at the project site.
Basin-fill margin	Alluvial fans at the edge of the salar	The pathway for diffuse recharge and the location of the density transition.
Basement	Depth unknown locally	Lower boundary of the flow system; a target for gravimetry.

3.2 Parameters

Two parameters govern the answers in Section 4. Transmissivity T controls how far and how fast the cone of depression propagates. Specific yield Sy controls how deep it goes. Neither has been measured here, so Section 4 is worked across the range each plausibly occupies.

PARAMETER	RANGE USED	BASIS FOR THE RANGE
Specific yield, Sy	0.02 to 0.20	Spans the values commonly reported for halite and sediment sequences in salars. Total porosity is higher and is not the relevant figure — much of the pore brine in a halite crust does not drain.
Transmissivity, T	100 to 5,000 m²/day	Spans from a poorly connected, compartmentalized sequence to a well-developed permeable horizon. Producing horizons in salars usually sit in the upper half of this range.
Brine density	approximately 1.21	Risacher and Rettig, consistent with the model's own volumetric conversion of 424 mg/L. Governs Section 4.5.
Extraction, Q	33,500 m³/day	Base case. The reduced-recovery case at 18.23 Mm³/yr is 49% higher and scales all drawdowns proportionally.

One constraint links the two parameters: a formation cannot at the same time have very low transmissivity and deliver 33,500 m³ per day. The low-T rows in Section 4 are shown for completeness, but a wellfield in such ground would fail to produce at the required rate long before the drawdowns in those rows developed. The realistic band is the upper part of the range.

3.3 The community supply and the San Gerónimo aquifer

Beneath the salar the pore fluid is brine. It is brine in the crust and brine at depth, across the whole interior, and the wellfield sits 18 km in from the edge. There is no freshwater aquifer beneath the salar and this assessment predicts no freshwater level, because there is none to predict.

The community of Río Grande is supplied with groundwater from the aquifer of the San Gerónimo basin — alluvial sediments — through wells of the order of 50 m deep in the Pucapampa sector and through springs fed by that same aquifer. There is no surface intake from the river and no municipal network. The community supply is therefore groundwater, and the question that would otherwise dominate this assessment is not settled by the source: it has to be answered.

TECHNICAL ANALYSIS

What links extraction to the San Gerónimo aquifer, and in which direction

The project extracts no freshwater: it takes brine from beneath the salar and produces its water by condensing what evaporates from it. There is no competition for the same resource. But the San Gerónimo aquifer and the brine beneath the salar are not bodies isolated from each other: they meet at a density transition on the basin margin, and that transition is precisely the connection.

The direction matters, and it runs in the community's favor. Lowering the BRINE table reduces the head of the dense fluid; the freshwater side is left with relatively higher head, so the freshwater advances toward the salar and the interface moves inward, not toward the community wells. The most common fear — that pumping brine pushes saltwater toward the community's wells — corresponds to the reverse mechanism and does not apply here. Section 4.5 quantifies that mobility: about 4.8 m of displacement per meter of change in hydraulic head, roughly eight times less mobile than the seawater interface it is instinctively compared with.

What remains open is the other side of that same coin: if the induced gradient causes the aquifer to drain marginally more toward the salar, that is a loss of freshwater from the basin the community drinks from, accumulated over 43 years. It is the same mechanism this assessment already identifies for the Río Grande de Lípez in its delta, now applied to the aquifer. It is expected to be second-order and small, but that must be measured and not assumed; Section 8 includes it. The project's 200 m³/day supply is, in any case, independent of that aquifer: it is produced from brine inside the plant. In a dry year that independence is worth more than the volume.

What does remain a genuine hydrogeological question is the dilution of the resource itself. The marginal inflows and the delta dilute the brine where they enter the salar, and the model's own sources place the richest lithium in the southeast, near the river mouth. If the cone of depression extends toward that dilution zone, the grade of the produced brine could fall over time. That is the project's exposure, not the community's, and Section 4.5 addresses it.

3.4 Recharge and discharge

- **Inflows:** direct precipitation of about 350 mm/yr on the salar surface, the Río Grande de Lípez and other marginal inflows, and diffuse recharge through the alluvial fans.
- **Outflows:** evaporation from the shallow brine table through the crust, which is the dominant natural discharge in a closed salar, plus lateral movement within the basin.
- The relationship between evaporation and brine-table depth over a salt crust is the most important term to measure. It controls how the natural system responds to a lowered brine table, and it cannot be transferred from other salars — the crust, the precipitation and the evaporation regime of Uyuni differ materially from those of Atacama.

TECHNICAL ANALYSIS

A feedback that works in the project's favor and must nonetheless be quantified

In a closed salar, evaporation from the shallow brine table is the main natural loss. When extraction lowers the brine table, that evaporative loss decreases. Part of the extracted volume is therefore offset by evaporation that no longer occurs, rather than coming entirely from storage.

This is a real and well-documented mechanism, and it reduces net depletion below the gross extraction figure. It cannot be invoked without measuring the evaporation–depth relationship, and the offset is not proportional — it saturates once the brine table falls below the depth at which capillary evaporation operates.

Section 4 ignores this offset throughout. Every figure it contains is therefore conservative — larger than a calibrated model incorporating the offset should produce.

SECTION 04

4. Anticipated effects of extraction

4.0 What it is that falls

Every drawdown figure in this section is a lowering of the brine table — the phreatic surface of a body of saturated brine of specific gravity close to 1.21, beneath the salar. Nowhere in this assessment is a freshwater level predicted, because there is no freshwater body beneath the salar for which to predict one.

These figures are a matter of resource sustainability, not a matter of water for the communities. The community of Río Grande is supplied from the San Gerónimo aquifer, a body distinct from that of the brine; Section 3.3 sets out in which direction they are related and what remains to be measured. The figures below concern whether the wellfield can sustain 43 years of production and whether the produced grade holds — they do not concern anyone's drinking water.

One further point of basis, which runs through this whole section. The wells are YLB's existing wells, so the drawdown assessed here is not additional to an existing cone — it is a change in the depth of a cone that already exists around a wellfield that has been pumping for years. The figures are given at 12.23 Mm³/yr, the project's own extraction, and at 29.35 Mm³/yr, the bound that governs if YLB simultaneously resumes full licensed extraction. Section 2.4 explains why both are shown.

A note on terminology, because the distinction is easily lost in translation and in summaries. "The water table falls 14 meters" would be an incorrect description of this result, and would be understood as a statement about drinking water. The correct statement is that the brine table beneath the salar falls of the order of 14 m near a wellfield sited 18 km into the salar, and that the river is unaffected. For that reason this document uses "brine table" throughout.

4.1 Method

Three standard analytical treatments are applied, each answering a different question, each worked across the parameter ranges of Section 3.2. Together they bound the answer enough to be useful, and they do so without a single borehole.

QUESTION	METHOD	RESULT IN
How much water leaves storage, and how deep is the resulting decline?	Volumetric: $h = V / (A \times Sy)$	Section 4.2
How far does the cone of depression reach?	Radius of influence: $r = \sqrt{(2.25\ T\ t / S)}$	Section 4.3
How much drawdown at a given distance?	Thiem steady state: $s(r) = Q / (2\pi T) \times \ln(R/r)$	Section 4.4
How far does the brine–freshwater interface move?	Density relation: $\Delta z = \Delta h \times \rho f / (\rho b - \rho f)$	Section 4.5

Common assumptions and their direction of error: homogeneous and isotropic aquifer, constant density in the flow solution, and no evaporation offset. The first two are simplifications; the third makes the results conservative. A fourth assumption must be stated in the opposite sense: the Thiem solution is steady-state and presupposes a constant-head boundary at radius R that supplies the flow in perpetuity. It cannot therefore be invoked as "no recharge to the cone". The use of the Cooper-Jacob radius inside the Thiem

equation is the quasi-steady approximation of common practice, and is to be understood as such here. Real heterogeneity will distort the cone, which is why these are bounds and not predictions.

4.2 Storage depletion

Extraction over the life of the project is 526 Mm³ at the project's own 12.23 Mm³/yr, and 1,262 Mm³ at the upper bound of 29.35 Mm³/yr. If that volume is taken from brine storage and the resulting decline is distributed uniformly over an area, the lowering of the brine table in meters is as follows. The table corresponds to 526 Mm³; every figure in it is multiplied by 2.40 at the upper bound.

AFFECTED AREA	SY = 0.02	SY = 0.05	SY = 0.10	SY = 0.20
250 km²	105.2 m	42.1 m	21.0 m	10.5 m
500 km²	52.6 m	21.0 m	10.5 m	5.3 m
1,000 km²	26.3 m	10.5 m	5.3 m	2.6 m
2,000 km²	13.1 m	5.3 m	2.6 m	1.3 m
5,000 km²	5.3 m	2.1 m	1.1 m	0.5 m
10,000 km² (whole salar)	2.6 m	1.1 m	0.5 m	0.3 m

Section 4.3 supplies the missing datum: the area over which the decline actually develops.

4.3 Radius of influence

After 43 years of continuous extraction, the outer limit of measurable drawdown, in kilometers:

TRANSMISSIVITY	SY = 0.02	SY = 0.05	SY = 0.10	SY = 0.20
100 m²/day	13.3	8.4	5.9	4.2
500 m²/day	29.7	18.8	13.3	9.4
1,000 m²/day	42.0	26.6	18.8	13.3
2,500 m²/day	66.5	42.0	29.7	21.0
5,000 m²/day	94.0	59.4	42.0	29.7

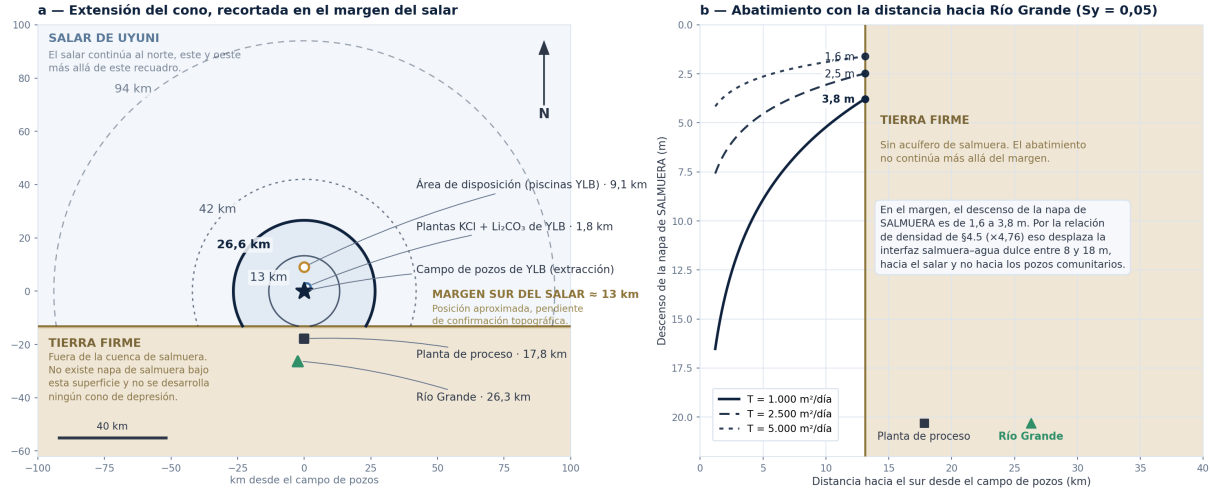
The cone reaches between about 4 and 94 km across the full parameter range; the central band — T of 1,000 to 2,500 m²/day with Sy = 0.05 — sits between 27 and 42 km. A radius of 13 km encloses some 530 km²; one of 40 km, some 5,000 km²; one of 94 km, some 27,700 km² — close to three times the surface area of the salar.

These figures are radii, not actual extents. The analytical solution assumes a laterally continuous and infinite aquifer. The brine aquifer is not: it ends at the salar margin, and to the south that margin lies about 13 km from the wellfield — that is, inside every radius in the table except the smallest. South of that line lie the process plant, at 17.8 km, and Río Grande, at 26.3 km, both on dry land: there is no brine table there, and therefore no cone of depression to draw. Figure 2 cuts the arcs off at that boundary rather than drawing them over ground where the aquifer does not exist. The 94 km extreme exceeds the physical basin in every direction and is retained only as an analytical bound.

What is determined in that direction is the drawdown at the margin itself: 1.6 to 3.8 m in the central transmissivity band, which by the density relation of Section 4.5 displaces the brine–freshwater interface by about 8 to 18 m, toward the interior of the salar. That is the magnitude that matters for the connection with the San Gerónimo aquifer (Section 3.3), and not the radius. Carrying the mid-range areas back to Section 4.2 gives declines of the order of 1 to 10 m. What the radius does not define is a population at risk — the cone is in brine, and there is no freshwater beneath the salar for it to reach.

Figura 2 — Extensión del cono de influencia y límite físico de la cuenca

Radios centrados en el campo de pozos de YLB. Planta a escala real, norte arriba. El cono existe únicamente donde existe salmuera.



El margen sur del salar se sitúa a unos 13 km del campo de pozos. Los círculos se interrumpen allí: la solución analítica supone un acuífero lateralmente continuo, y el acuífero de salmuera termina en el margen. Al sur de esa línea — donde se encuentran la planta de proceso y Río Grande — no hay napa de salmuera y no se desarrolla ningún cono de depresión. El radio de 94 km excede la cuenca física en todas las direcciones y se conserva únicamente como límite analítico, no como extensión prevista. Los descensos representados son de la napa de SALMUERA en el interior del salar; no son niveles de agua dulce. Secciones 4.3, 4.4 y 4.5.

Figure 2 — Extent of the cone of influence and physical boundary of the basin, centered on YLB's wellfield. **(a)** The arcs stop at the southern margin of the salar: the analytical solution assumes a laterally continuous aquifer, and the brine aquifer ends there. South of the margin — where the process plant and Río Grande are located — there is no brine table and no cone develops. **(b)** Drawdown with distance in that same direction, with Sy = 0.05. The circles enclose a lowering of the BRINE table beneath the salar, not of any freshwater body. The position of the southern margin was estimated from satellite imagery (Esri / Maxar) and requires topographic confirmation; see Section 8.1.

TECHNICAL ANALYSIS

What actually determines the cone, and why the radius is not the measure

The cumulative interaction is not a question of radius. The project extracts from YLB's wells, so both extractions share a single cone by construction — at any transmissivity in the table, including the most pessimistic. The cumulative-impact chapter of the EEIA must therefore be built on the combined extraction from a single wellfield, and cannot be satisfied by reference to YLB's own filing.

Grade over time. If the cone extends toward the delta and the marginal inflows, diluted brine is drawn toward the wellfield and the produced grade drifts downward over 43 years. That is the project's own exposure and it compounds with the grade uncertainty of Section 2.2.

The shallow regime. The brine table beneath this salar sits at or very near the surface — the project's own geotechnical note describes trench excavation in a wet salt crust with brine at the surface. Lowering it by several meters over a cone tens of kilometers across is a change to the shallow system, not just to storage at depth. Whether that alters the morphology of the crust or the depth and duration of seasonal flooding appears nowhere and is not a question this assessment can answer. It matters directly to the project: the conveyor piles, the pipeline trench and the disposal area are all founded in or on that crust, and all were designed against its current behavior.

Where the piezometers go. A network confined to the wellfield measures the wrong thing. It must extend along the cone, to the margin, and to the boundaries with the other operators' areas.

4.4 Drawdown with distance

Steady-state drawdown of the brine table as a function of distance from the center of the wellfield, with Sy = 0.05. The table corresponds to the project's own extraction of 12.23 Mm³/yr. Dashes indicate distances beyond the radius of influence for that case. These are brine levels in the interior of the salar, not freshwater levels at the margin.

T (M²/DAY)	R (KM)	2 KM	5 KM	10 KM	20 KM
100	8.4	76.6 m	27.7 m	—	—
500	18.8	23.9 m	14.1 m	6.7 m	—
1,000	26.6	13.8 m	8.9 m	5.2 m	1.5 m
2,500	42.0	6.5 m	4.5 m	3.1 m	1.6 m
5,000	59.4	3.6 m	2.6 m	1.9 m	1.2 m

- In the realistic band — transmissivity of 1,000 m²/day or higher — drawdown is of the order of 4 to 14 m near the wellfield, 3 to 9 m at 5 km, and 1 to 5 m between 10 and 20 km, with the project's own extraction. At the upper bound of 29.35 Mm³/yr every figure is multiplied by 2.40, as set out below.

DISTANCE FROM WELLFIELD	AT 12.23 MM³/YR — PROJECT ONLY	AT 29.35 MM³/YR — UPPER BOUND
Near the wellfield (2 km)	4 to 14 m	9 to 33 m
5 km	3 to 9 m	6 to 21 m
10 to 20 km	1 to 5 m	2 to 12 m

- The 100 m²/day row shows 77 m at 2 km, but that case cancels itself out: a formation of that transmissivity could not sustain 33,500 m³/day. It is retained to show that the failure mode in poor ground is production failure, not silent over-extraction.
- The radius of influence does not change with the extraction rate — it is governed by transmissivity, storage and time, and the range of 13 to 94 km holds in every case. Only the depth of the cone scales with pumping. Recovery remains the single largest lever the project itself controls, and it is a process parameter, not a hydrogeological one.
- Density is not part of the Thiem solution, but it is part of transmissivity: $T = (k \cdot \rho \cdot g/\mu) \cdot b$. The brine of this salar has a density of 1.21 and a viscosity of the order of two to three times that of freshwater at Uyuni temperatures, so the ratio ρ/μ is roughly half that of freshwater. For the same intrinsic permeability, transmissivity to brine is therefore lower and drawdown greater, not smaller. The range of 100 to 5,000 m²/day used here comes from the literature and has not been measured on this brine; establishing which fluid it corresponds to is part of the program in Section 8.3.

4.5 Dilution of the resource

Where the marginal inflows enter the salar they dilute the brine, and the transition between diluted brine and full-strength brine is a density boundary. Its mobility in response to a change in hydraulic head is given by the density relation:

$$\Delta z = \Delta h \times \rho f / (\rho b - \rho f)$$

Density relation and displacement of the hydraulic interface

With a brine of specific gravity close to 1.21 and freshwater at 1.00, that ratio is 4.76. In a coastal aquifer, where seawater is only 2.5% denser than freshwater, the same ratio is 40.

CHANGE IN HYDRAULIC HEAD	INTERFACE DISPLACEMENT HERE	SAME CHANGE, COASTAL AQUIFER
1 m	4.8 m	40 m
2 m	9.5 m	80 m
5 m	23.8 m	200 m
10 m	47.6 m	400 m

TECHNICAL ANALYSIS

The result

The brine–freshwater interface at Uyuni is approximately eight times less mobile per meter of change in hydraulic head than the seawater interface it is instinctively compared with. The very high density contrast that makes this brine difficult to process is the same property that makes its interface stable.

This does not mean the interface can be ignored. A displacement of tens of meters is still a displacement, it accumulates over 43 years, and it bears both on the grade of the produced brine and on any freshwater body it approaches. But the coastal intuition — that a few meters of drawdown drag an interface hundreds of meters — is wrong here by almost an order of magnitude.

Saline intrusion is among the concerns most frequently raised about brine extraction, and it is reasonable to raise it. The answer here is more favorable than the coastal analogy suggests, and it rests on a physical property of this brine and not on any operating practice — which is why anyone with the density figures can verify it independently.

4.6 What these results do and do not establish

- **They establish magnitude and extent:** a cone of influence most plausibly of 13 to 94 km, a drawdown of a few meters up to about fourteen near the wellfield, and an interface that moves approximately five meters for every meter of change in hydraulic head.
- **They say nothing about the community supply, and now they need to.** Río Grande is supplied from the San Gerónimo aquifer through wells and springs; the direction of the relationship is set out in Section 3.3, and what must be measured, in Section 8.
- The connections worth quantifying run in the reverse direction: how much additional water the Río Grande de Lípez loses to infiltration in its delta, and how much the San Gerónimo aquifer drains toward the salar, once the brine table falls. The first is measured against the river's flow and requires a gauging record; the second requires piezometry at the margin and in the aquifer itself. Both are expected to be small — but they are the mechanisms that link extraction to the water the community actually uses, and they should be closed rather than assumed.
- **They do not capture heterogeneity.** A low-permeability barrier between the wellfield and the community changes the local answer entirely, in either direction.

- **They do not capture the time dimension.** Thiem is steady-state; the transient approach to it over 43 years determines when effects appear, not whether they do.
- **They will be replaced by a calibrated variable-density numerical model.** Their function is to establish now whether this project sits in a range where careful design suffices, or in one where the concept itself is in question. On these figures, it is the former.

SECTION 05

5. Project water balance

5.1 The balance

The project produces its process water by condensing what evaporates from the brine. No fresh groundwater and no community water is extracted. The balance below corresponds to base-case extraction; the water terms scale with it.

TERM	VOLUME	NOTE
Brine extracted	12.23 Mm³/yr	The only water the project takes from the environment, and it is brine, not freshwater.
Condensate recovered	5.38 Mm³/yr	14,740 m³/day. Produced by bringing the brine to salt saturation by mechanical means.
Process consumption	−2.40 Mm³/yr	6,575 m³/day, at a rate of 30 m³ per tonne of product.
SURPLUS	**2.98 Mm³/yr**	8,165 m³/day available before any committed withdrawal.
Workers' camp	−0.100 Mm³/yr	273.7 m³/day for 1,400 people at 195.5 L/person/day.
Community supply to Río Grande	−0.073 Mm³/yr	200 m³/day, the committed minimum.
Wet cleaning of the solar field	−0.055 Mm³/yr	About 150 m³/day across the photovoltaic and solar thermal fields, 18 to 24 cleanings per year.
UNCOMMITTED SURPLUS	**≈ 2.75 Mm³/yr**	Approximately 7,540 m³/day with no assigned destination.
Make-up water required in operation	0	Structural, not conditional. The plant condenses 5.38 Mm³/yr and consumes 2.40; it produces more water than it uses by virtue of the flowsheet itself. Operating losses — water of crystallization in the carnallite, residual moisture in the solids, water in the magnesium reject liquor — come out of the water that enters with the feed brine, not from an external source. The external-source requirement corresponds to construction and to the pilot-plant years, before steady-state condensate exists; see §5.2.

5.2 The claim of no freshwater use

The claim holds, and it holds with a margin that is worth stating clearly: everything the project has committed to supply — camp, community and panel washing together — amounts to about 624 m³/day against a surplus of 8,165 m³/day. The committed withdrawals consume less than 8% of the water the plant produces as a by-product.

It is not a marginal claim that requires careful wording. It is a structural feature of the flowsheet: concentrating brine by mechanical means necessarily produces distillate, and the process needs less of it than the concentration stage yields.

The constraint that does bind today on this salar. YLB operating reports indicate that the potassium and lithium plants currently running at Uyuni operate below nominal capacity for lack of water, and warn that water could cap production before energy does. It is the only documented cause of the fact already recorded in Section 2.2 — that YLB has not reached its nominal capacity with this brine. That constraint does not carry over to this project, and the reason is one of flowsheet and not of merit: a pond route evaporates to the atmosphere and must replace the evaporated water; the pond-free route evaporates mechanically and condenses what it evaporates, so it produces water instead of consuming it. The contrast is worth keeping in mind when comparing this project with those already operating on the same salar, and it is not a siting advantage: any operation on this salar that replaced ponds with mechanical evaporation would obtain the same result. The same observation is developed, from the energy-supply side, in Section 6.2 of the companion document.

TECHNICAL ANALYSIS

Three conditions on which the claim depends

Process demand of 30 m³/t. The model notes that 50 m³/t is the figure for a plant without wash-water recirculation. At 50 the balance remains positive — the surplus falls from 2.98 to 1.4 Mm³/yr, that is from 8,165 to 3,781 m³/day, and to 3,157 m³/day once all committed withdrawals are deducted (§5.4) — so in that case too no external source is required in operation. What changes is not sufficiency but margin. The claim should be stated against a demand figure that the process EPC guarantees.

Construction. The balance above corresponds to operation. Construction water — concrete, compaction, dust suppression, hydrostatic testing of two pipelines — is required before any condensate exists, because what is being built is precisely the plant that produces the water. This is the only stage in which the project could actually need an external source, and it has not been quantified anywhere in the model.

The pilot-plant years. The 10,000 t/yr pilot plant operates from year 1. Its condensate is proportionally smaller, while the commitments to the camp and the community are not. The balance should be run separately for the pilot phase.

5.3 Continuity of the community supply

Volume is not the governing constraint on the 200 m³/day commitment — it amounts to 2.4% of the surplus. Continuity is, because the water is a by-product of an operating plant and ceases when the plant stops.

TERM	VALUE	BASIS
Committed supply	200 m³/day	The contractual minimum.
Existing process reserve	13,150 m³	Two days of process demand at 6,575 m³/day. Sized to process need, not to the commitment.
Storage for a four-day interruption	800 m³	The multi-day cloud event against which the emergency generation is designed. At minimum load the plant is not concentrating brine and produces no condensate.
Storage for 30 days of autonomy	6,000 m³	Fully decouples the commitment from plant availability.

- The existing reserve is process storage. It is not allocated to the community supply and its size responds to process demand, so it does not scale with the commitment.
- The recovered water is essentially distilled and is not potable without remineralization to the WHO calcium and magnesium targets. That plant is a precondition of the commitment, not an accessory.
- Dedicated storage, sized in days of autonomy and not only by daily flow, is what makes the supply independent of plant availability. At 200 m³/day the volumes involved are insignificant against the scale of the project.

5.4 The uncommitted surplus

There is one water commitment: 200 m³/day to the community of Río Grande, contractual for 43 years. In addition, the project will establish a collection point at the plant, open to all the communities of Nor Lípez and supplied from that same surplus; Section 5.4.1 describes what that requires physically. The contractual volume commitment remains the 200 m³/day to Río Grande. What follows quantifies what remains after that commitment and the plant's own needs, so that the volume is on record and can be discussed openly rather than discovered later.

	BASE CASE, 30 M³/T	PESSIMISTIC CASE, 50 M³/T
Condensate produced	5.38 Mm³/yr	5.38 Mm³/yr
Process consumption	2.40 Mm³/yr	4.00 Mm³/yr
Surplus	2.98 Mm³/yr — 8,165 m³/day	1.38 Mm³/yr — 3,781 m³/day
Less camp, solar cleaning and the Río Grande commitment	−624 m³/day	−624 m³/day
UNCOMMITTED	**7,541 m³/day**	**3,157 m³/day**

The pessimistic column is the one to use as the working basis. A process demand of 30 m³/t is an assumption the EPC has not guaranteed; 50 m³/t is what a plant without wash-water recirculation consumes. Even at 50 m³/t, of the order of 3,150 m³/day remain surplus to every current requirement, including the Río Grande commitment.

How that surplus is shared among the communities is not a technical question and is not decided in this document. The figure is recorded here so that CUPCONL has it.

5.4.1 Making it available without building a network

The surplus is a by-product with no assigned destination. It will be made available at a collection point at the plant, open to all the communities of Nor Lípez, which requires no transmission infrastructure. The contractual volume commitment remains the 200 m³/day to Río Grande; the collection point is one of access, and its flow is set by the available surplus.

- The water is already produced and already remineralized for the camp and for the supply to Río Grande. A collection point draws on the same treated stream.
- What it requires physically: a filling standpipe, storage sized to the withdrawal rate and not to demand, and a protocol on who may collect and when. No pipeline and no per-capita distribution: the point delivers to whoever arrives, under a protocol agreed with the communities.
- Storage at the collection point scales with the flow withdrawn, not with the available surplus. At a withdrawal of 200 m³/day, two days of autonomy is 400 m³; at 1,000 m³/day, it is 2,000 m³.
- Remineralization capacity scales with the water actually delivered, not with the water produced. Treating only what leaves the site keeps that plant small.

The distinction that matters technically: delivering water through a pipeline network is a capital project whose cost is set by distance and by the dispersion of the settlements. Delivering it at a collection point is not — the marginal cost of an additional cubic meter collected is close to zero, because the water is produced whether or not anyone withdraws it.

One point worth putting on record for the EEIA. This water is produced from brine inside the plant, so it is independent of precipitation and of river flow. Condensate production peaks in the dry season, which is when the Río Grande de Lípez is at its lowest. The surplus is therefore greatest at the time of year when it would be worth most.

SECTION 06

6. Hydrogeology of the reject streams

Two streams leave the plant and require a destination: some 3.67 Mt/yr of solids and 1.467 Mm³/yr of magnesium reject brine. Both are governed by extraction and both scale with it.

6.1 Solid rejects on the surface of the salar

The design intent is explicit in the model: on a salar, the disposal site is the salar. The halite returns to the crust — inert, the same material the crust is made of — within a delimited YLB disposal area located about 30 km from the plant, reached by the 38 km conveyor. Nothing is placed on dry land, and nothing is built as a landform. That transforms the analysis from a question of impoundment capacity into a question of area, thickness and what a salar that floods every year does with the material once placed.

6.1.1 Area, and what constrains thickness

With 88 to 113 Mm³ of material over 43 years, the required area is determined by the layer thickness the crust can support:

PLACEMENT THICKNESS	AT 88 MM³	AT 99 MM³	AT 113 MM³	LOAD
1 m	88 km²	99 km²	113 km²	16 kPa
2 m	44 km²	50 km²	57 km²	31 kPa
3 m	29 km²	33 km²	38 km²	47 kPa
5 m	18 km²	20 km²	23 km²	78 kPa
10 m	9 km²	10 km²	11 km²	157 kPa

The constraint is bearing capacity, not embankment design. The crust is a few meters of halite over brine-saturated lacustrine mud. It is not a foundation capable of supporting ten meters of salt: the project's own geotechnical note already warns of salt creep, dissolution and seasonal flooding for structures far lighter than this, and specifies 6 to 12 m piles for a conveyor. A salt load of 157 kPa over several square kilometers will settle, and could punch through into the underlying mud.

Realistic placement is therefore shallow — plausibly one to three meters — which puts the required area at approximately 30 to 100 km². For scale, that is equivalent to between three and ten percent of the entire salar, and to between five and sixteen times the photovoltaic field. Whether the designated disposal area comes anywhere near that size appears in no available source.

TECHNICAL ANALYSIS

The first thing to establish, and it requires no drilling

How large is the designated disposal area, and what thickness of salt will the crust support there? Those two figures together determine whether 43 years of production have somewhere to put their solids.

Bearing capacity is answered with plate load tests and a few shallow boreholes in the crust and the underlying mud — a small, inexpensive campaign that can be run at the same time as the survey of the area.

The regulatory framework does not supply the figure. DS 4180 recognizes residual salt deposits as an instrument in its Article 15 and refers their design to the environmental mining regulations in force, but sets neither a placement thickness nor a minimum bearing capacity on salt crust. That has to be established by testing.

Until both are known, the claim of "zero cell-construction capex" rests on nothing. It may well turn out to be correct. It has simply never been verified against 158 million tonnes.

6.1.2 What the salar does with the material

The argument that salt is inert holds for halite. It does not hold for the stream as a whole.

STREAM	OVER 43 YEARS	BEHAVIOR ON A SALAR THAT FLOODS ANNUALLY
Halite, NaCl	112 Mt	Genuinely inert. The same material as the crust, and it dissolves and recrystallizes with the seasonal cycle without altering the chemistry of what it sits on.
Carnallite, $KMgCl_3 \cdot 6H_2O$	33 Mt	Deliquescent and highly soluble. Absorbs atmospheric moisture, releases magnesium and potassium chloride liquor, and dissolves readily when the salar floods.
Magnesium sulfate	13 Mt	Highly soluble. Mobilized by floodwater.
Combined non-inert fraction	**46 Mt — 29% of the stream**	This is the part that does not simply become crust again.

Approximately 29% of what is placed on the salar is soluble magnesium-bearing salt, and the salar floods four months a year. Every rainy season a fraction of that dissolves into standing water that remains on the surface and moves across it. Where it then goes, and at what concentration, is not a question of infiltration — it is a question of surface redistribution, and it is not addressed in any of the sources.

- Establish what proportion of the carnallite and the magnesium sulfate dissolves per rainy season, and where the resulting liquor goes — laterally across the surface, or downward into the crust.
- Sample brine and standing water upstream and downstream of the disposal area before the first tonne arrives. Without that baseline the project will later be unable to distinguish its own contribution from YLB's existing liability at the same site.
- Assess whether repeated seasonal dissolution and recrystallization cements the deposit, changes its permeability, or alters the surface hydrology of the area it occupies.
- Assess salt dust generation in the dry season, when the surface is loose. Dust is the pathway by which a disposal area on an open salar reaches communities and solar fields tens of kilometers downwind.
- Segregate the streams if the assessment supports it. Halite genuinely can return to the crust; the magnesium salts are a different material with a different destination, and disposing of them together forfeits the one part of this waste stream that has a clean answer.

That last point is worth weighing early. "We return the salt to the salt" is defensible for 112 Mt of halite and indefensible for 46 Mt of soluble magnesium salts mixed into it. Separating them turns one hard disposal problem into an easy one plus a smaller hard one — and it is far simpler to build into the design now than to modify after an EEIA has been filed on the combined stream.

6.2 The magnesium reject brine

1.467 Mm³/yr of magnesium chloride reject liquor — 4,020 m³/day, and 63 Mm³ over the life of the project — leave the plant as a stream conveyed by a separate pipeline and return underground by injection. No surface dam or pond forms part of this project.

Injection wells in permeable formations typically accept between several hundred and a few thousand cubic meters per day each.

INJECTION RATE PER WELL	WELLS REQUIRED	NOTE
500 m³/day	8	Conservative — a low-permeability or scaling-prone target.
1,000 m³/day	4	—
1,500 m³/day	3	—
2,000 m³/day	2	Optimistic; a very permeable and compatible horizon.

With redundancy and allowance for wells lost to clogging over 43 years, a working program is of the order of five to twelve wells with rolling replacement. Compared with two to ten square kilometers of geomembrane (1.467 Mm³/yr at between 0.75 and 0.15 m/yr of net evaporation as declared in §9.3), the comparison is not even close.

There is also a symmetry worth pointing out. The project withdraws 12.23 Mm³/yr from the formation and would return 1.47 Mm³/yr to it — around 12% of what it takes, re-entered into the same basin. That does not cancel the drawdown of Section 4, but it is not nothing either, and it is the kind of thing that should be quantified in the eventual numerical model rather than left as a footnote.

6.2.1 The question that actually has to be answered

Not whether to reinject, but where. The returned magnesium chloride must not reach the production zone, or it would raise the Mg/Li ratio in the brine the plant is trying to process — the project would be poisoning its own feed. The location of injection is therefore a hydrogeological and a process question at once:

- A target horizon that accepts the volume, is hydraulically separated from the producing horizon, and does not communicate with it over 43 years.
- Sufficient distance and gradient separation from the wellfield so that the plume does not migrate toward it. Injection and extraction in the same basin are not independent operations, and the same numerical model must contain both.
- Compatibility with the formation. The reject brine is scaling-prone and the receiving formation contains its own brine; the incompatibility that precipitates in the well is the usual failure mode of injection schemes and is diagnosed with laboratory mixing tests before any well is drilled.
- Injectivity and fracture pressure, from step-rate and pressure fall-off tests, with a maximum allowable injection pressure set below the formation fracturing pressure, with margin.

- Well integrity over 43 years in corrosive service, plus allowance for workovers and an abandonment specification.

TECHNICAL ANALYSIS

The real obstacle is regulatory, not technical

An instrument exists, and it is not written for this stream. The Regulation for the Integrated Management of Waste from the Evaporite Sector (RGIRSE), approved by Supreme Decree 4180 of 12 March 2020, establishes in its Article 15 that "brines may be reinjected into the salar if they are of equal or better quality than the well brine and present no contaminants, subject to prior laboratory analysis and the respective supporting technical report approved by the AAC". It is the only written pathway, and its wording corresponds to the return of unprocessed brine.

Two concrete mismatches. The reject liquor is depleted in lithium and concentrated in magnesium relative to the well brine, so it does not satisfy the "equal or better quality" condition on a literal reading. And Article 15 speaks of reinjection *into the salar*; what this section proposes is injection into a target horizon hydraulically separated from the producing horizon, which is not the same thing and which the text does not contemplate.

What still does not exist. Neither a construction and integrity standard for injection wells, nor a regulated maximum allowable pressure, nor a defined monitoring requirement. The regulation provides an approval pathway but not a technical standard to design against.

This must be raised early with the MMAyA and the AAC, and resolved before the EEIA is filed — and not discovered during review. It is the only point on the waste side that cannot be resolved by engineering, and it has a long lead time to resolution.

A pilot injection program, carried out before committing, is what turns the proposal from a modeling argument into demonstrated performance. It is also the material from which the supporting technical report that Article 15 itself requires is built, before a regulator that has no precedent whatsoever to work from.

SECTION 07

7. Findings

NO.	FINDING	CONFIDENCE AND BASIS
1	Base-case extraction is 12.23 Mm³/yr — 33,500 m³/day — and the working range across plausible grades and recoveries is 8 to 27 Mm³/yr.	High. Arithmetic from production, grade and recovery. The range, and not the base figure, is the honest statement.
2	The project opens no new wellfield. Brine is extracted from YLB's existing wells, within a volume already licensed and already assessed. No new point of extraction and no new area of influence is created on this salar. Total extraction from those wells is 12.23 Mm³/yr if the project's extraction replaces YLB's, and 29.35 Mm³/yr if it adds to it in full.	High as to structure. The distinction between the two cases cannot be pinned down without YLB's pumping records, which is why Section 4 contains both.
3	The project is a net producer of water. Condensate exceeds process demand by 8,165 m³/day in the base case and by 3,781 m³/day with a pessimistic process demand. All committed supplies together consume less than 8% of the base-case surplus. The water constraint that today limits YLB's plants on this same salar does not carry over to this project — see §5.2.	High. Structural, not marginal — the flowsheet cannot concentrate brine without producing distillate.
3B	Between 3,150 and 7,540 m³/day are surplus to every current requirement, including the Río Grande commitment. It will be made available at a collection point at the plant, open to all the communities of Nor Lípez, which requires no transmission infrastructure. The only contractual volume commitment remains the 200 m³/day to Río Grande.	High as to volume. A collection point is a filling standpipe, storage sized to the withdrawal rate, and a protocol. Whether it is established is a matter of agreement, not a technical conclusion.
4	Construction-phase water demand is not quantified in any of the sources, and it is the only stage in which the project could need an external supply.	High confidence that the gap exists; as yet no basis to estimate its magnitude.
5	The community supply is secure in volume and exposed in continuity. It depends on an operating plant, and the planned storage is sized for the process, not for the commitment.	High. Dedicated 30-day storage, of about 6,000 m³, would eliminate the exposure.
6	The BRINE table falls of the order of 4 to 14 m near the wellfield, 3 to 9 m at 5 km and 1 to 5 m between 10 and 20 km with the project's own extraction — and 2.40 times those figures at the upper bound of 29.35 Mm³/yr. The cone most plausibly reaches 13 to 94 km in every case, because the radius does not scale with the pumping rate; the upper end of that range exceeds the physical basin and lies outside the validity range of the method. These are brine levels in the interior of the salar, not freshwater levels in the communities.	Medium-high. The radius of Section 4.3 and the volumetric depletion of Section 4.2 span the full parameter range of Section 3.2; the drawdown with distance of Section 4.4 is calculated at Sy = 0.05 and does not span that range. They ignore the evaporation offset, so they are conservative in that term. They substitute for a numerical model only for as long as none exists.
6B	The brine–freshwater interface moves about 4.8 m per meter of change in hydraulic head — roughly eight times less than coastal-aquifer intuition suggests, because the density contrast is eight times greater.	High as to the relation; the current position of the interface remains unknown and requires geophysics.
6C	There is no freshwater beneath the salar: the pore fluid is brine throughout the interior and the wellfield sits 18 km in from the edge. The community of Río Grande is supplied from the San Gerónimo aquifer through wells of about 50 m and springs. They are distinct bodies, but not isolated: they meet at a density transition on the margin. The links are second-order and operate in the reverse direction — a lower brine table causes the river to lose somewhat more to infiltration in its delta and the aquifer to drain somewhat more toward the salar. Neither pushes brine toward the community wells.	Medium. The DIRECTION of the effect is firm and rests on the density contrast of Section 4.5, verifiable by anyone with the density figures. The MAGNITUDE is not: the induced drainage from the San Gerónimo aquifer has been neither measured nor estimated. Until it is, this finding establishes the sense of the effect, not its size.
7	The solids are placed on the surface of the salar, not on dry land. At a thickness the crust can reasonably support — one to three meters — 43 years of production require approximately 30 to 100 km², that is, three to ten percent of the salar. Neither the area of the designated site nor the bearing capacity of the crust there appears anywhere.	High as to volume and arithmetic. Both constraints are inexpensive to establish: a survey of the area and plate load tests, with no drilling campaign needed.
7B	Around 29% of the stream — 46 Mt of carnallite and magnesium sulfate over the life of the project — is soluble and deliquescent, and the salar floods four months a year. "Returning the salt to the salt" is true for the 112 Mt of halite and not for the rest. Seasonal dissolution and surface redistribution are not addressed in any source.	High. Segregation of halite from the magnesium salts should be assessed now, while it is a design decision and not a later modification.

NO.	FINDING	CONFIDENCE AND BASIS
8	The reject brine returns underground: 1.467 Mm³/yr through some five to twelve injection wells with rolling replacement. No surface dam or pond forms part of this project.	High as to the conclusion; medium as to the number of wells.
9	The open questions on reinjection are location and licensing, not feasibility. The returned magnesium must not reach the production zone or it would raise the Mg/Li ratio in the plant feed. DS 4180 provides in its Article 15 for the reinjection of brines into the salar subject to AAC approval, but conditions it on a quality equal to or better than that of the well brine — a condition this liquor does not satisfy on a literal reading — and sets no standard for injection wells and no monitoring requirement.	High as to the regulatory text, cited in §9.2. The pathway exists but is not written for this stream; resolving it with the MMAyA and the AAC has a long lead time and should begin before any field work.
9B	The shallow consequences of lowering a brine table that today sits at the surface have not been examined. Both the morphology of the crust and the depth and duration of seasonal flooding depend on the shallow regime, and the conveyor piles, the pipeline trench and the disposal area are all founded in or on that crust.	Stated as a gap. The structures were designed against the current behavior of the crust, over a cone tens of kilometers across and 43 years of drawdown.
10	Grade and recovery remain the two largest levers the project controls over its own hydrogeological footprint — extraction, drawdown, solids and reject brine all scale with them. Neither has been measured for this project, but the wells are YLB's and have been producing, so their records are the fastest route to both.	High. This is the argument for running the confirmation drilling and the hydraulic characterization as a single campaign.

SECTION 08

8. What must be measured, and in what order

The findings above fall into three groups: those that can be closed at the desk, those that require a season of measurements, and those that require drilling. They should be addressed in that order, because the inexpensive ones change the design questions that the expensive ones must answer.

8.1 Can begin now, without field work

Compilation of YLB's own wellfield records — flow rates and pumping histories, water levels, brine analyses and any test data from the wells this project will use. These wells have been in operation for years. This is now the highest-value item in the entire program: it turns assumed parameters into measured parameters without a meter of new drilling, and resolves the extraction question of Section 2.4 that Section 4 currently carries as a range. That such records exist in usable form and are available is an assumption of this assessment, and establishing what YLB actually holds is itself the first task.

- Survey of the area of YLB's designated disposal site, plus plate load tests and shallow boreholes in the crust and the underlying mud to establish what thickness of salt it will support. Together they close Finding 7. Inexpensive, fast, and with no drilling campaign needed.
- Construction-phase water demand, from the construction plan. Closes Finding 4.
- Pilot-phase water balance, run separately from the operating balance. Closes part of Finding 3.
- Sizing of storage for the Río Grande commitment, expressed in days of autonomy and not only by daily flow. Closes Finding 5.
- Compilation and critical review of the existing hydrogeological data on the salar, including YLB's drilling, piezometry and brine analyses. Establishes what the field program does not need to repeat.

8.2 A season of measurements

- Open with the MMAyA and the AAC the question of injection licensing, specifying how Article 15 of DS 4180 applies to a magnesium chloride reject liquor and what supporting technical report will be required. No field work, with a long lead time to resolution. Closes the obstacle in Finding 9.
- Evaporation from the salt crust as a function of brine-table depth, measured on site. Governs the natural discharge term and therefore the evaporation offset that makes Finding 6 conservative.
- Weather stations at the plant, the wellfield and the basin margin, recording from the earliest possible date, given that a full hydrological year is the minimum useful record.
- Extent of rainy-season flooding and its relationship to brine-table depth, from multi-temporal imagery and the piezometric record. Serves the conveyor corridor and the disposal area, and begins to answer whether extraction can alter the seasonal flooding against which those structures were designed.

8.3 Requires drilling and geophysics

- Wellfield grade and hydraulic parameters, from YLB's own wells. Take YLB's records first; drill only what those records do not already answer. Any new work should be run as a single campaign together with the hydraulic program.
- Specific yield, measured on preserved core by unit and by interval, and checked against pumping tests. The parameter that decides whether Finding 6 sits at one meter or at forty.

- Long-duration pumping tests, of no less than thirty days, with a full observation network. What distinguishes a laterally continuous aquifer from a compartmentalized one.
- Position of the brine/freshwater interface, by transient electromagnetics and resistivity profiling calibrated against multilevel piezometers.
- Gauging of the Río Grande de Lípez over a full hydrological year, with an estimate of infiltration loss in the delta. This establishes the baseline of the community's supply and closes the one mechanism that links extraction to the river.
- Injection target horizon and its confining units; laboratory mixing tests for formation compatibility; injectivity and pressure fall-off tests; and a pilot injection program before committing. The location must be fixed jointly with process engineering, so that the returned magnesium cannot reach the production zone.

TECHNICAL ANALYSIS

The point about sequence

The reject brine evaporation test and the survey of the disposal area together cost a small fraction of a single borehole, and between them they decide the disposal strategy — which in turn decides how much drilling is needed and where.

Running them first is not caution. It is the difference between drilling a program designed around a known disposal route and drilling one that must be extended when the route changes.

SECTION 09

9. Basis and sources

9.1 Project sources used

SOURCE	STATUS	WHAT WAS TAKEN FROM IT
Techno-economic model — GEA & Grade, rows 250 onward	Current design basis	Extraction, brine per tonne, grade variable, recovery, evaporation load, condensate, process water balance, brine pipeline specification.
Techno-economic model — Magnesium & Reagents, rows 40–56 and 68–131	Current	Route C chemistry, salt yields, reject brine fraction, solids tonnages, conveyor routing and distance.
Techno-economic model — Power; Firm Supply (Diesel)	Current	Solar field capacity, battery sizing, emergency generation, photovoltaic field stages.
Techno-economic model — Camp & Community Water	Current	Camp demand and per-capita rates. The 200 m³/day community supply is a confirmed commitment and is accounted for in full in this assessment. It currently shows as zero in the cost model and must be reinstated there; this is an accounting correction, not a change to the commitment.
Site master plan	Current	Component footprints.
Techno-economic model — Water Recovery; GEA & Grade rows 1–249; Magnesium & Reagents rows 1–39 and 57–67	WITHDRAWN — not used	Nothing. These sections describe the superseded flowsheets, with ponds and bulk liming. They are listed here so that it is on record that they were identified and excluded.

9.2 Published sources

- YLB environmental filings, EEIA-AI KCl (2012) and EEIA-AI Li_2CO_3 (2020) — brine flow of 17.12 Mm³/yr and freshwater of 1.22 Mm³/yr for 350,000 t of KCl and 15,000 t of Li_2CO_3.
- Regulation for the Integrated Management of Waste from the Evaporite Sector (RGIRSE), Supreme Decree 4180 of 12 March 2020 — Article 15, storage of industrial waste from the processes of exploitation and concentration of salts and brines, and the conditions under which brines may be reinjected into the salar subject to AAC approval.
- CEDIB — pond inventory from the field visit of December 2023, brine intensity of 1,231 m³ per tonne of Li_2CO_3, average lithium grade of the salar of 250 ppm, and documented lithium recovery below 55%.
- USGS — lithium-rich brines of the Salar de Uyuni, from 0.34 to 2.03 g/L depending on location, richest in the southeast.
- Peer-reviewed literature on lithium recovery from magnesium-rich salt-lake brines — the Route C flowsheet and its basis of 68 m³/t without losses, Uyuni Mg/Li ratio of 20 ± 2, lithium exceeding 2,000 ppm in the south in Nor Lípez, salt yields per 400 m³ of brine, and the winter and summer temperature ranges.
- Energy report "Potosí, Bolivia: gas and electricity situation and the energy demand of a lithium plant" (July 2026) — water constraint on YLB's operating plants at Uyuni, attributed there to operating reports. It is

the published source of the contrast noted in Section 5.2; the rest of that report — national grid, gas and energy intensity — is the subject of the energy-supply document and is not used here.

- Risacher and Rettig (1980) — composition of the brine of the central salar, 349 mg/kg lithium, basis of the volumetric mean of 424 mg/L.

9.3 Assumptions introduced in this assessment

The following do not come from the model. They are stated so that they can be challenged.

ASSUMPTION	VALUE USED	WHY
Specific yield range	0.02 to 0.20	Spans the values commonly reported for halite and sediment sequences in salars. Presented as a range precisely because no value has been measured here.
Area over which drawdown develops	250 to 10,000 km²	Spans from a strongly compartmentalized aquifer to the whole salar. Not a prediction.
Bulk density of the solids	1.4 to 1.8 t/m³	Range for crystallized salt rubble. Determines the disposal volume.
Net evaporation of the reject brine	0.15 to 0.75 m/yr	The upper bound is the model's salar brine rate; the lower bounds reflect that concentrated $MgCl_2$ is deliquescent. Must be measured.
Injection rate per well	500 to 2,000 m³/day	Generic range for industrial injection wells. It is formation-specific and is not verified here.
Project life for volumes	43 years of production	Per the model. Note that the sources also contain figures of 25, 45 and 50 years for different purposes.

Comments

This assessment is delivered to be examined. Comments from the technical advisers designated by CUPCONL may be addressed in writing to Heitaria Swiss AG, Rigistrasse 1, 6374 Buochs, Switzerland, and will be answered in writing.

Where a comment bears on a parameter, Section 9.3 identifies which are open to substitution, and the arithmetic of Sections 2 and 4 is laid out in detail so that another value can be substituted and the consequence followed without redoing the document.

ANNEX A

What European battery buyers will ask about water

From 18 August 2027, every company that places batteries for electric vehicles, light transport or industrial use on the European Union market must carry out due diligence on the lithium in them and have it verified by an independent notified body (Regulation (EU) 2023/1542, Articles 47–53; date as amended by Regulation (EU) 2025/1561). Annex X of the Regulation lists water among the environmental risks the buyer must examine, together with soil, biodiversity and human health. In brine lithium, water is the question buyers ask first, because the evaporation-pond route consumes it.

This annex adds nothing to the hydrogeology. It shows where the answers a buyer's auditor will look for already sit in this assessment, and which are still open. Construction water, the drainage of the San Gerónimo aquifer, the soluble fraction of the solids and the reinjection licence are the points on which an auditor would press; they are listed as open for the same reason as in Section 7 — so that they are read as open, not as omitted. Closing them in the order of Section 8 is what turns this assessment into evidence a buyer can rely on.

A.1 The questions, and where the answers are

WHAT THE BUYER WILL ASK	WHAT THIS ASSESSMENT ESTABLISHES	STATUS AND WHERE
Does the plant take fresh water from the territory?	No, in operation. The plant condenses 5.38 Mm³/yr from the brine and consumes 2.40; the surplus is 8,165 m³/day, and every committed supply together uses less than 8% of it. Make-up water in operation is zero by construction of the flowsheet.	**High.** §5.1, §5.2; Finding 3.
And during construction and the pilot years?	Construction water is not quantified in any source, and it is the only stage at which an external supply could be needed. The pilot-plant balance has not been run separately.	**Open.** §5.2; Finding 4.
Is the communities' drinking water secure?	The 200 m³/day to Río Grande is 2.4% of the surplus and secure in volume. It is exposed in continuity because it depends on an operating plant; dedicated storage of about 6,000 m³ (30 days) removes the exposure. The water requires remineralisation before it is potable.	**High** as to volume; exposure stated. §5.3; Finding 5.
What does brine extraction do to the aquifers?	No new wellfield: brine comes from YLB's existing wells. The brine table falls about 4 to 14 m near the wellfield and 1 to 5 m at 10 to 20 km. These are brine levels inside the salar, not freshwater levels in the communities. The links to the San Gerónimo aquifer run in the reverse direction and are second-order; their magnitude has not been measured.	**Medium-high** as to magnitude; **open** as to the San Gerónimo drainage. §4; Findings 2, 6, 6C.
What happens to the salar surface and the rejects?	No ponds and no surface dam. The reject brine is reinjected underground. The solids occupy roughly 30 to 100 km² of salar surface over 43 years; about 29% of them are soluble magnesium salts, and their seasonal dissolution in the flooded months is not addressed in any source. The reinjection route under DS 4180 still has to be resolved with the MMAyA and the AAC.	**Open** on soluble solids and licensing. §6; Findings 7, 7B, 8, 9, 9B.